

Project Charter – Flood Pattern Discovery Using Copernicus Sentinel Data

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1. Charter Introduction

This Project Charter formally authorizes the Flood Pattern Discovery Project, a semester-long academic initiative conducted under the Data Analytics program. The project applies Earth observation data engineering and supervised machine learning techniques to historical satellite datasets in order to identify, analyze, and interpret flood-related spatial and temporal patterns.

The project is explicitly historical in scope. It examines flood events occurring between June and August 2024 in Sirajganj District, Bangladesh, with particular emphasis on the major flood event of 12 July 2024 along the Jamuna River. While executed during the current academic term, the analysis focuses entirely on retrospective satellite data.

Flooding in Bangladesh represents a persistent systemic hazard driven by monsoon rainfall, transboundary river systems, low-lying topography, and increasing climatic variability. Sirajganj District, situated along the Jamuna River floodplain, is especially vulnerable to seasonal inundation that disrupts livelihoods, damages infrastructure, and displaces communities. By grounding the analysis in a real-world disaster context, the project balances analytical rigor with geographic specificity while maintaining feasibility within a single academic semester.

The project leverages open-access satellite data from the Copernicus Sentinel program. Sentinel-1 Synthetic Aperture Radar (SAR) imagery provides cloud-penetrating, all-weather surface observations, while Sentinel-2 optical imagery provides complementary reflectance-based indicators of water, vegetation, and land cover. The integration of SAR and optical datasets enables more robust flood detection than either modality alone.

The project aligns with the Government of Canada National Project Management System (NPMS) principles by clearly defining scope, governance, assumptions, risks, and deliverables. It also aligns conceptually with international disaster risk reduction priorities articulated by the United Nations Office for Disaster Risk Reduction (UNDRR), emphasizing scalable, reproducible analytical workflows rather than operational forecasting systems.

1.1 Document Change Control

This document is subject to controlled updates in accordance with structured project governance practices. Substantive changes to scope, objectives, methodology, or deliverables require formal revision tracking and sponsor acknowledgment.

Minor editorial or formatting changes that do not alter scope or intent do not require re-approval.

Revision History:

- Version 1.0 – Initial Draft
- Version 1.1 – Expanded academic framing and deliverables
- Version 1.3 – Transition from unsupervised to supervised learning methodology
- Version 1.4 – Technical consolidation, validation refinement, quantitative estimates, and distinction-level enhancements

1.2 Executive Summary

Flooding remains one of the most destructive natural hazards globally, particularly in riverine and delta regions where hydrological variability intersects with dense population and infrastructure vulnerability. Bangladesh experiences recurrent flood events that challenge disaster response systems and long-term development planning.

Advances in satellite Earth observation provide continuous, near-global surface monitoring capability. Copernicus Sentinel-1 SAR data enables reliable water detection independent of cloud cover, while Sentinel-2 optical imagery provides high-resolution reflectance data suitable for computing water and vegetation indices.

This project integrates Sentinel-1 and Sentinel-2 datasets accessed via the Copernicus Data Space Ecosystem (CDSE) using a catalog-first STAC-based workflow. Historical observations from June–August 2024 are processed within a defined floodplain corridor AOI in Sirajganj District.

A supervised learning approach is used to classify flooded vs. non-flooded surfaces. Training labels are derived from authoritative external reference products where available, including Copernicus Emergency Management Service (EMS) rapid mapping and UNOSAT flood extent datasets. Permanent water masks from JRC Global Surface Water are used to distinguish baseline water from event-driven inundation.

The analytical workflow emphasizes transparency, reproducibility, and structured validation. Final outputs include flood classification rasters, aggregated flood-area statistics, and Power BI reporting artifacts suitable for academic evaluation and policy-oriented review.

1.3 Authorization

This Charter authorizes the Flood Pattern Discovery Project and grants the project team the authority to apply academic, computational, and analytical resources in accordance with the defined scope and methodology.

Any material change to scope or analytical approach requires sponsor review.

Project Sponsor: George Campanis
Project Lead: Samrat Chakraborty

2. Project Overview

The project is structured as an applied satellite data analytics initiative integrating data engineering, feature extraction, supervised modeling, validation, and reporting.

The design balances analytical ambition with semester-level feasibility, ensuring all stages—from acquisition through reporting—can be completed to a high standard within available academic resources.

2.1 Project Summary

The project consists of five structured phases:

1. Data acquisition
2. Preprocessing and feature engineering
3. Exploratory analysis
4. Supervised modeling
5. Reporting and visualization

Each phase builds logically upon the previous stage, culminating in validated flood classification outputs.

2.1.1 Project Goals, Business Outcomes, and Objectives

Goals

- Demonstrate effective use of Earth observation data for flood pattern analysis.
- Develop a reproducible and transferable analytical workflow.

Objectives

- Acquire and preprocess Sentinel-1 and Sentinel-2 data for the defined AOI.
- Engineer meaningful SAR and optical features.
- Train and validate supervised models for flood detection.
- Produce structured analytical documentation and visualization outputs.

Business Outcomes

- Improved understanding of flood dynamics in the study region.
- Transferable analytical methodology for other flood-prone regions.
- Academic and policy-aligned insights supporting disaster risk reduction frameworks.

2.1.2 Scope Definition

In Scope

- Historical data extraction (June–August 2024).
- Sentinel-1 GRD (VV+VH) and Sentinel-2 Level-2A processing.
- Feature engineering and supervised classification.
- External validation and quantitative evaluation.
- Reporting and visualization deliverables.

Out of Scope

- Real-time monitoring.
- Flood forecasting or predictive modeling of future events.
- Operational deployment or alert systems.

2.1.3 Boundaries

The project is limited to a single AOI within Sirajganj District to ensure analytical depth and feasibility. No additional geographic regions will be processed during this term.

Field validation and ground-truth data collection are outside scope. Validation relies on satellite-derived external reference datasets.

2.2 Milestones

Key milestones include:

- Completion of data acquisition inventory
- Validation of preprocessing workflows
- Feature stack construction
- Supervised model training
- Validation metric reporting
- Final analytical documentation

Detailed scheduling is maintained in the Gantt Chart appendix.

2.3 Deliverables

- Project Charter
- Entity Relationship Diagram (metadata/results registry)
- Data Dictionary
- Power BI Analytical Report

- Final Analytical Report
- Gantt Chart

All deliverables support transparency, evaluability, and reproducibility.

2.4 Technical Approach

2.4.1 Data Acquisition Architecture

All satellite data is accessed via the Copernicus Data Space Ecosystem using its STAC API.

Workflow:

1. Query Sentinel-1 and Sentinel-2 collections using AOI and date filters.
2. Filter by orbit direction, polarization, cloud cover, and processing level.
3. Generate inventory of available scenes.
4. Selectively download required assets.

Authentication uses CDSE OAuth2 tokens with automated refresh handling and retry logic.

2.4.2 Area of Interest (AOI)

The AOI is defined as a WGS84 GeoJSON floodplain corridor buffered along the Jamuna River.

Initial bounding box:

Lat: 24.01638–24.785471

Lon: 89.253643–89.819781

Estimated AOI area: 3,000–5,000 km².

2.4.3 Preprocessing Pipeline

Sentinel-1 (SAR)

Apply Orbit File → Thermal Noise Removal → Radiometric Calibration (σ^0) → Speckle Filtering → Terrain Correction (DEM-based) → dB conversion.

Dual polarization (VV+VH) used. Orbit direction retained as metadata.

Sentinel-2 (Optical)

- Level-2A reflectance used.
- Cloud masking via Scene Classification Layer (SCL) and/or s2cloudless.
- Bands resampled to common 10 m grid.

Multi-sensor alignment performed via nearest-in-time pairing.

2.4.4 Feature Engineering

SAR Features

- $\sigma^{\circ}VV$
- $\sigma^{\circ}VH$
- VV/VH ratio
- Temporal backscatter change

Optical Features

- NDWI
- MNDWI or AWEI
- NDVI
- NIR and SWIR reflectance

Temporal Features

- Pre-event composite
- Event composite
- Rolling statistics or variability metrics

2.4.5 Supervised Learning Methodology

Primary task: Binary classification (flood vs non-flood).

Baseline model: Random Forest classifier.

Additional exploration: Gradient Boosted Trees if feasible.

Training labels derived from:

- Copernicus EMS rapid mapping
- UNOSAT flood extent
- JRC Global Surface Water baseline

Model selection justified based on interpretability, robustness, and compute feasibility.

2.4.6 Validation Strategy

Spatially stratified train/validation splits are used to reduce spatial leakage.

Metrics:

- Precision
- Recall
- F1 Score
- Intersection-over-Union (IoU)
- Confusion Matrix

External flood extent datasets used for quantitative overlap comparison.

2.5 Data Infrastructure and Reproducibility

- Version-controlled Git repository
- requirements.txt or environment.yml
- Structured directory hierarchy
- Cloud-Optimized GeoTIFF (COG) rasters
- GeoJSON vector outputs
- CSV/Parquet aggregated reporting outputs

Power BI integration uses zonal statistics, flooded area time series, and percent flooded metrics.

Model artifacts and evaluation reports archived for full computational traceability.

2.6 Risks, Assumptions, and Constraints

Risks

- Cloud cover interference
- SAR speckle noise
- Temporal misalignment of labels
- Wet soil misclassification

Mitigation

- Temporal compositing
- Cross-sensor feature fusion
- Structured validation against authoritative references

Assumptions

- Continued availability of Copernicus data
- Sufficient academic compute resources

Constraints

- Fixed academic semester timeline
- Limited learner cloud environment capacity

3. Project Organization

The project follows a lean governance structure appropriate for an academic environment. The Project Lead oversees execution, while the Sponsor provides milestone-level oversight.

4. Project References

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